We'll we'll get going. And I think we're all here because we are very interested in foundation models and we don't need to spend much time saying what they are or describing. They know we're all fascinated by these models that across domains from single cell biology, histology to sequence to function are able to potentially solve many different tasks. And that's one. So there we go. So yeah, we'll start when I put together these slides to kind of kick off the session. And we're going to start saying we are all here. We'll get going. And I think we're all here because we are very interested in foundation models and we don't need to spend much time saying what they are or describing. I know we are all fascinated by these models that across domains from single cell biology to histology to sequence to function are able to potentially solve many different tasks. And that's one. That's one. So there we go. So yeah, Gustavo and I put together these slides to kind of kick off the session. And we're going to start. So I was saying we are all here because we are very interested in when and how we can use this foundation model for different tasks. And this, of course, requires benchmarking, an assessment. But the question that we want to address here today, and at least start thinking as a community, and really welcome all of you to engage and contribute, is what type of benchmark we should be doing for these foundation models? Is it just the usual one, or do we need a different kind of benchmark? And we know that the classic benchmarking has been very successful and very helpful for computational biology. We heard yesterday John Jumper describing how CASP was critical in the development of AlphaFold. And in that type of benchmark, regardless of the domain area, what we do is to have some data, some training data to build models. We typically keep withhold some data, which is the test data, that we use to assess the value of certain models for a certain task based on some type of scoring. And we know this is a helpful way acting to clinical trials as a double-blind manner to assess the value of models, because those that assess the models are not those that develop them. Therefore, the assessment is unbiased. And if we ask the model developers to send the actual model with the code, we can look at it, we can look at all the details, and we can rerun it. So we improve reproducibility. So this concept, as I said, has been applied in many contexts. CAS for protein structures. Gustavo and I and many others in the room are part of DREAM. And there are many other initiatives like CAMTAC, Haji, and so forth. So in principle, we can say, okay, we will use this paradigm to assess our foundation models, and we look at specific tasks in each of the domains, and we'll see if the foundation models, or perhaps fine-tune foundation models, are able to then perform one specific task. And that's helpful, that's something that's already ongoing, but it doesn't seem like it's giving us really the whole picture of what these foundation models do, because they are able to do, in principle, any task. And the tasks that we choose are typically driven by kind of current trends, what the community is interested in, as well as by the technical limitations that we have. And this may lead to some type of street light effect bias, right? That we are just looking at some areas that are of interest, but not really capturing the value of these models as a whole. I have this one. You can use it. Okay. So, and thinking about this, it feels a bit like when the

first explorers went, let's say, to America and they started to delineate the map of the continent, and they had like a cartography which was incomplete. Some areas, like the Caribbean area, where the first one were in high resolution. Other areas, let's say, South America and the Amazonas, at some point, this is now from the 17th century, they were starting to understand. But other areas, like northwest or Alaska, they had no clue what's there. So by analogy, then in our case in computer biology, as we heard yesterday in the case of structural biology, we really understand very well how to solve this problem. Other things such as virtual cells, perturbation on cells, is very active at the moment. There are many initiatives ongoing. But it could be that there are other areas that we don't even know what they are. So then the question that we want to discuss today is how many of which tasks are enough to give us a good cartography of these models, of these foundational models. Because it's not also going to be that we're going to be able to do any possible experiment and simulate or generate every possible data. That would be similar to the tale of Vorgas that you may know called Rigorant Science, which he describes this fantastical world where cartographers were trying to make the best possible map, but at the end of the map was as big and including everything in the actual kingdom. So that's kind of also not what we can achieve. So we try to find a balance between covering areas that are important, but they are enough for us to generalize about the value of models. So how can we go there? Just some ideas. And again, this is food for thought and for discussion. We think the different benchmarking activities across domains, like CAS, Green, CAMDA, KACHI, and many others, should come together and share expertise and experiences. We also should engage the community, both experimentalists who can generate the data that we need, as well as the modelers, those of us developing models, to formulate the questions in the data, to sample this kind of territory, this cartography of tasks. And going beyond this and going beyond the current paradigm, we think we should think of new approaches and how we can leverage new strategies, potentially aggregating micro-tasks, some type of community science or crowdsourcing that allow us to benchmark what's happening. And this is a bit similar, and this is a slide from Gustavo about what happens when you go to ChatGPT. You may have seen that when you ask a question, it lets you the option to say, I like it, I don't like it, or sometimes it may give you actual answers for you to choose what is, in your eyes, the better one. So this is a way how we as users give feedback to the tool, and this allows them to improve its ability. So inspired by this concept, Nicolas, who is here, since still with Gustavo and Gustavo, developed an exercise, what we're going to do today, whereby we are now transcribing with this little thing here, our talks. You are going to be able to ask us questions, and I'll see in a moment the system, about the talks. Then we will pass on to two large language models. Then they will give us answers, and then also we will assess the value of this answer to see how well this works. This is something we will do as persons, but that also does not scale up. So we are starting to think, and some people even in our group at ABI, together with Sebastian, who may be here somewhere, we are thinking of how to use actual agents and language models to do the benchmarking. But for now, this

is an exercise. We're going to be the agents evaluating the results. Just to give you a sense of what it will look like, so we will give you the moment the URL where you can go. You'll be able to pick a talk. You'll be able to ask questions like, what did Julio, which Borgerstell did Julio mention today? Then we will ask the LMs, we will get the answers, and then you will be able to evaluate the answers. But we need a ground truth. In our case, the ground truth against which we will assess the answers is going to be the actual speakers, which hopefully they know what they talk about. And we'll use that to see how well we do and to build a leaderboard as we usually do in this crowdsourcing metrics. So that's the idea for today. Please stay. We have great scientific talks. You'll hear from Bob Bank, from Toronto, Maria Birbik, Pablo Meyer, Liz Fassbender, Ansul Kundager, Justin and I. You'll hear the talks, you'll learn a lot of great science, and then you'll be able to participate in this experiment with us where we try to see how well we can crowdsource the assessment of models. And then we'll finish, you don't see this, but we'll finish with a panel discussion about these topics. So to conclude this relatively short introduction, so what we want to kind of discuss today is new approaches that we think are needed to benchmark foundation models. We think the solutions could include a consensus of tasks and also crowdsourcing of this macro task, as we will see in this pilot that we will use an early implementation of this idea. And the larger vision that we have is that as a community, we develop models and we also generate data. Some of this data we'll use it to generate ground truths that in turn through different tasks will allow us to benchmark the models and kind of pass or not pass models and then develop repositories of curated models. There are activities at EBI to do such a curated model repository, also CCI is developing them. And then these curated models can then be used again by the community to analyze data sets and generate insights. So with that, I'd like to finish the introduction. Thanking especially Nicolas, who is here. You see, this is what he looked like yesterday. He's somewhere here, who set it up. All the Dream Directors, many are here, Pablo, Luca, Gustavo, Kyle, for really great input into this. Our sponsors, Embod BI, Chanz Zuckenberg, Tempus, and IBM Research. On the right, you can see the URL or the QR code already going to the system. So the username is going to be audience and the password audience pass. We also have So we wrote in a white box because we don't necessarily agree with black box models, we brought a white box with a series here of guidelines of how we can do this. A handful is for the speakers, for the grand truth, is I will not give you. But we can pass this along. And it should be clear, but maybe it will help you to participate. I think that's it, Nagu Sabo. So thanks again. Stay and enjoy. And if Bo is around, is Bo one? We know he's here, or we've told he's in Liverpool. No, Bo Wang. Okay, Maria, I think that's the we can we have two or three minutes for questions, so anybody wants to ask questions to Julio on these ideas? Okay. Okay, I hope all some of you will come at five for this exercise. And those of you who stay, please ask the questions that will allow us to see whether this crowdsourcing of microtasks is feasible. If not, boys, Maria here. Okay. Maria? I saw Maria because we are very interested in when and how we can use this foundation model for different tasks. And this, of course, requires benchmarking, an assessment. But the question that we want to address here today, and at least start thinking as a community, and really welcome all of you to engage and contribute, is what type of benchmark we should be doing for these foundation models? Is it just a usual one, or do we need a different kind of benchmark? And we know that in auto-classic benchmarking has been very successful and very helpful for computational biology. We heard yesterday John Jamper describing how CASP was critical in developing the FalphaFold. And in that type of benchmark, regardless of the domain area, what we do is to have some data, some training data to build models. We typically keep withhold some data, which is test data, that we use to assess the value of certain models for a certain task based on some type of scoring. And we know this is a helpful way acting to clinical trials as a double-blind manner to assess the value of models, because those that assess the models are not those that develop them. Therefore, the assessment is unbiased. And if we ask the model developers to send the actual model with the code, we can look at it, we can look at all the details, and we can rerun it. So we improve reproducibility. So this concept, as I said, has been applied in many contexts. CAS for protein structures. Gustavo and I and many others in the room are part of DREAM. And there are many other initiatives like CAMTAC, Hiji, and so forth. So in principle, we can say, okay, we will use this paradigm to assess our foundation models, and we will look at specific tasks in each of the domains, and we'll see if the foundation models, or perhaps fine-tune foundation models, are able to then perform one specific task. And that's helpful, that something's already ongoing, but it doesn't seem like it's giving us really the whole picture of what these foundation models do, because they are able to do, in principle, any task. And the tasks that we choose are typically driven by kind of current trends, what the community is interested in, as well as by the technical limitations that we have. And this may lead to some type of strict right effect bias, right? That we are just looking at some areas that are of interest, but not really capturing the value of these models as a whole. I have this one. Okay. So, and thinking about this, it feels a bit like when the first explorers went, let's say, to America and they started to delineate the map of the continent, and they had like a cartography which was incomplete. Some areas, like the Caribbean area, where the first went, were in high resolution. Other areas, let's say, South America and the Amazonas, at some point, this is now from the 17th century, they were starting to understand. But other areas, like northwest or Alaska, they had no clue what's there. 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discussion about these topics. So to conclude this relatively short introduction, so what we want to kind of discuss today is new approaches that we think are needed to build Smart Foundation models. We think the solutions could include a consensus of tasks and also crowdsourcing of this micro task, as we will see in this pilot that we will use on every implementation of this idea. And the larger vision that we have is that as a community we develop models and we also generate data. Some of this data will use it to generate ground truths that in turn through different tasks will allow us to benchmark the models and kind of pass or not pass models and then develop repositories of curated models. There are activities at EBI to build such a curated model repository, also CCI is developing them. And then these curated models can then be used again by the community to analyze data sets and generate insights. So with that, I'd like to finish the introduction. Thanking especially Nicolas, who is here. You see, this is what he looked like yesterday. He's almost here, there he is, who set it up. All the Dream Directors, many are here. Pablo, Luca, Gustavo, Kyle, for really great input into this. Our sponsors, MOLEBI, Chance Zuckerberg, Tempus, IBM Research. On the right, you can see the URL or the QR code already going to the system. So the username is going to be audience and the password audience pass. We also have... So we brought in a white box, because we don't necessarily agree with black box models, we brought a white box with a series here of guidelines of how we can do this. A handful is for the speakers, for the ground truth, is I will not give you. But we can pass this along. And it should be clear, but maybe it will help you to participate. I think that's it, Naguiso. So thanks again. Stay and enjoy. And if Bo is around, we know he's here, or we've told he's in Liverpool. No, Bo Wand. Okay. Marie, I think that's the we can we have two minutes for questions, so anybody wants to ask questions to Julio on these ideas. Okay. Okay, I hope all some of you will come at at five for this exercise. Uh and those of you who stay, please ask the questions that will allow us to see whether this crowdsourcing of micro tasks is is feasible. If not, boys, Maria here. Okay. Maria? I saw Maria.